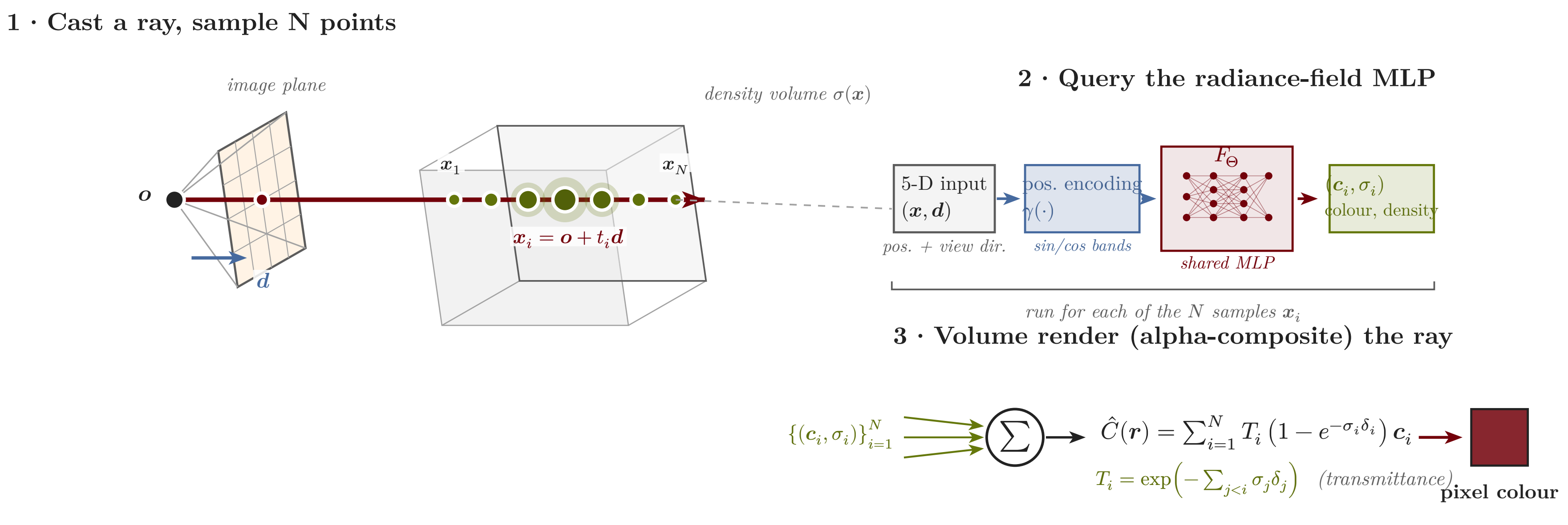
1 · Cast a ray, sample N points
image plane
density volume $\sigma(\boldsymbol{x})$
$\boldsymbol{o}$
$\boldsymbol{d}$
$\boldsymbol{x}_1$
$\boldsymbol{x}_N$
$\boldsymbol{x}_i = \boldsymbol{o} + t_i\boldsymbol{d}$
2 · Query the radiance-field MLP
5-D input $(\boldsymbol{x}, \boldsymbol{d})$
pos. + view dir.
pos. encoding $\gamma(\cdot)$
sin/cos bands
F_Θ
shared MLP
$(\boldsymbol{c}_i, \sigma_i)$ colour, density
run for each of the N samples $\boldsymbol{x}_i$
3 · Volume render (alpha-composite) the ray
$\{(\boldsymbol{c}_i, \sigma_i)\}_{i=1}^{N}$
$\sum$
$\hat{C}(\boldsymbol{r}) = \sum_{i=1}^{N} T_i \left(1 - e^{-\sigma_i\delta_i}\right) \boldsymbol{c}_i$
$T_i = \exp\left(-\sum_{j<i} \sigma_j\delta_j\right)$ (transmittance)
pixel colour